

Each question is worth 5 points for a total of 40 points. The Extra Credit problem can get you up to an additional 5 points.

Problem 1

Find the volume of the solid generated by revolving the region bound by the curves $y = x^2$, $y = 0$, and $x = 3$ about the x -axis.

Problem 2

Find the volume of the solid generated by revolving the region bound by the curves $y = 2x - 1$, $y = \sqrt{x}$, and $x = 0$ about the y -axis.

Problem 3

Find the lengths of the curve $y = \frac{x^3}{3} + \frac{1}{4x}$ with $1 \leq x \leq 3$

Problem 4

Find the area of the surface generated by revolving the curve $y = \frac{x^3}{9}$, with $0 \leq x \leq 2$, about the x -axis.

Problem 5

Let $f(x) = \frac{x+3}{x-2}$. Find $f^{-1}(x)$, the domains of $f(x)$ and $f^{-1}(x)$, and the ranges of $f(x)$ and $f^{-1}(x)$. Then find $f(f^{-1}(x))$ and $f^{-1}(f(x))$.

Problem 6

Solve

$$\int \frac{2y}{y^2 - 25} dy.$$

Problem 7

Solve

$$\int_{\ln(4)}^{\ln(9)} e^{x/2} dx.$$

Problem 8

Solve the differential equation

$$2\sqrt{xy}\frac{dy}{dx} = 1.$$

Extra Credit:

A colony of bacteria is grown under ideal conditions in a laboratory so that the population increases exponentially with time. At the end of 3 hours there are 10,000 bacteria. At the end of 5 hours there are 40,000. How many bacteria were present initially?